

1. Select all of the expressions that are equivalent to $(-8)^{-4} \cdot (-8)^2$.

☐ $\frac{1}{64}$

☐ $(-8)^{-8}$

☐ $(-8)^{-2}$

☐ $\frac{(-8)(-8)(-8)(-8)}{(-8)(-8)}$

☐ $\frac{(-8)(-8)}{(-8)(-8)(-8)(-8)}$

☐ $\frac{1}{(-8)(-8)(-8)(-8)(-8)(-8)(-8)(-8)}$

2. Select all of the expressions that are equivalent to $\left(\frac{1}{6}\right)^{-1}$.

☐ $-1\left(\frac{1}{2 \cdot 3}\right)^1$

☐ $-\frac{1}{6}$

☐ $\left(\frac{6}{1}\right)^1$

☐ $\frac{1^{-1}}{6^{-1}}$

☐ 6

Rewrite each expression in the form b^x , where x is a positive number.

3. $(-5)^{-5} \cdot (-5)^9$	4. $\left(\frac{12}{7}\right)^{-3}$
b^x where x is a positive number $(-5)^4$	b^x where x is a positive number $\left(\frac{7}{12}\right)^3$
5. $\left(\left(\frac{6}{11}\right)^6\right)^{-4}$	6. $\frac{13^{-8}}{13^{-2}}$
b^x where x is a positive number $\left(\frac{11}{6}\right)^{24}$	b^x where x is a positive number $\left(\frac{1}{13}\right)^6$
7. $(-18)^{-1}$	8. $(37)^{-24} \cdot (37)^{13}$
b^x where x is a positive number $-\frac{1}{18}$	b^x where x is a positive number $\left(\frac{1}{37}\right)^{11}$

9. Which expression is equivalent to $(y)^{-1}$?

A. $-1 \cdot y$

B. $-\frac{1}{y}$

C. y

D. $\frac{1}{y}$

Determine the value of each expression. If necessary, use the workspace to show your work.

10. $\frac{(1.1)^{-2} \cdot (1.1)^3}{(1.1)^3 \cdot (1.1)^{-5}}$

value

1.331

11. $\left(\frac{-8}{15}\right)^{-3} \cdot \frac{15^{-4}}{(-8)^{-2}}$

value

$-\frac{1}{120}$

12. $\left(\frac{15}{8}\right)^0 \div \left(\frac{2}{5}\right)^{-3}$

value

$\frac{8}{125}$

13. $\left(\left(\frac{9}{13}\right)^{-2}\right)^{-3} \cdot \left(\frac{9}{13}\right)^{-8}$

value

$\frac{169}{81}$

For each expression, apply exponent laws to write an equivalent expression with positive exponents that has the fewest factors possible. If necessary, use the workspace to show your work.

14. $(m^8 m^3)^2$

**equivalent
expression**

$$m^{22}$$

15. $x^4 \div ((x^2)^3)$

**equivalent
expression**

$$\frac{1}{x^2}$$

16. $\frac{y^5 y^{-8}}{y^{-3} y^2}$

**equivalent
expression**

$$\frac{1}{y^2}$$

17. $\left(\frac{2}{w^5} \cdot w^3\right)^{-4}$

**equivalent
expression**

$$\frac{w^8}{16}$$

18. $(4x^4y^5) \cdot (7x^{-8}y^6)$

19. $\left(\frac{a^{-5}n^{13}}{a^{-12}n^8}\right)^4$

**equivalent
expression**

$$\frac{28y^{11}}{x^4}$$

**equivalent
expression**

$$a^{28}n^{20}$$

20. Select all of the expressions that are equivalent to $\left(\frac{x^{-13}}{x^{11}} \cdot x^{-5}\right)^{-2}$.

☐ x^{14}

☐ x^{20}

☐ $\left(\frac{x^{-13}}{x^{11}} \cdot x^{10}\right)$

☒ $\left(\frac{x^{26}}{x^{-22}} \cdot x^{10}\right)$

☒ $\left(\frac{1}{x^{24}} \cdot x^{-5}\right)^{-2}$

☐ $(x^{-2} \cdot x^{-5})^{-2}$